Modeling Results and Observations:

To forecast loan default risk, a deep learning model was created using Keras with a TensorFlow backend following data preprocessing, feature transformation, and exploratory data analysis.To guarantee consistent model performance, the dataset was divided into training, validation, and testing sets, and numerical characteristics were standardized.The final model classified loan repayment outcomes correctly in most cases, with an accuracy of about 83% on the test dataset.The model's loss value of roughly 0.42 indicates that predictions were made without significant errors and with a decent degree of confidence.

The model's stability was demonstrated by the results' consistency over several runs.From a commercial standpoint, this model offers a trustworthy choice and a tool to help lenders detect higher-risk